Cardiovascular System (CARD) · NPLEX-P1-CARD-05

CV-5 Cardiac Muscle Contraction

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-05-01-C01				
NPLEX-P1-CARD-05-01-C02				
NPLEX-P1-CARD-05-01-C03				
NPLEX-P1-CARD-05-01-C04				
NPLEX-P1-CARD-05-01-C05				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-05-01-C01-BD1

Draw the ventricular action potential

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C01 Activity: NPLEX-P1-CARD-05-01-C01-BD1 Date: _____

Without looking at your notes, draw the action potential of a ventricular myocyte with voltage on the vertical axis and time on the horizontal axis. Label phase 4, phase 0, phase 1, phase 2, and phase 3. Beside each phase write the current and the named channel that dominates it, the direction that current carries charge, and the part of the ECG that phase corresponds to. Write the resting voltage, the threshold voltage, the peak voltage, how long the plateau lasts, and how long the whole action potential lasts. Then write what inward rectification means for the IK1 channel and give two things that property does for a working myocyte. Finish by writing how current reaches this cell in the first place and which structure in the intercalated disc carries it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-05-01-C01-BD2

Refractoriness and a change in plasma potassium

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C01 Activity: NPLEX-P1-CARD-05-01-C01-BD2 Date: _____

Write how long the effective refractory period lasts, the voltage at which it ends, what keeps the cell unexcitable during it, and how long the relative refractory period lasts and what a stimulus during it produces. Use that to explain in three or four sentences why the ventricle cannot be tetanized when skeletal muscle can. Next, a patient has a plasma K+ of 8.0 mmol/L (reference range 3.5 to 5.0 mmol/L). Work through the prediction in steps: calculate the new K+ equilibrium potential using 140 mmol/L inside the cell, write where the resting potential moves and why IK1 makes it move there, write what that does to the fast Na+ channels between beats, and write what happens to phase 0, to conduction velocity, to the QRS, and to phase 3 and the T wave. Say which ECG change appears first. Then write what a plasma K+ of 2.6 mmol/L does to phase 3 instead, and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-05-01-C02-BD1

Draw the dyad and the path from voltage to force

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C02 Activity: NPLEX-P1-CARD-05-01-C02-BD1 Date: _____

Draw one ventricular myocyte in cross section with a T tubule meeting a terminal cisterna of the sarcoplasmic reticulum. Label the T tubule, the dyad, the width of the cleft, the L-type Ca2+ channel with its protein name, the RyR2 cluster, the sarcoplasmic reticulum, and calsequestrin. Write where along the sarcomere the T tubule enters. Beside the drawing write the diastolic cytosolic free Ca2+, the peak cytosolic Ca2+ during a beat, the extracellular ionized Ca2+, and the share of activator Ca2+ that comes from the sarcoplasmic reticulum in human ventricle. Then write out calcium-induced calcium release in your own words, including what a Ca2+ spark is, what the Ca2+ transient is, and what ends release. Finish by writing the cross-bridge cycle from Ca2+ binding troponin C to the head detaching, naming each step and the point at which ATP is needed.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-05-01-C02-BD2

Remove the trigger, then let the store leak

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C02 Activity: NPLEX-P1-CARD-05-01-C02-BD2 Date: _____

An isolated heart and an isolated skeletal muscle are both moved into a Ca2+ free solution and stimulated. Write what happens to contraction in each over the next minute, how long each takes to change, and the structural reason for the difference. Say which transporter runs the cardiac store down once trigger entry stops. Then write what a drug that blocks cardiac L-type Ca2+ channels does to force and why skeletal twitches are spared. Next, a young person has a RyR2 mutation that lets the sarcoplasmic reticulum leak Ca2+ in diastole, and collapses during exercise. Write the sequence from adrenergic stress to the arrhythmia, naming what the Na+/Ca2+ exchanger does with the leaked Ca2+, the charge it moves per cycle, the resulting membrane event, and the name of that condition. Finish by naming two other situations in which the notes describe the same mechanism.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-05-01-C02-RT01

Rebuild excitation-contraction coupling

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C02 Activity: NPLEX-P1-CARD-05-01-C02-RT01 Date: _____

Rebuild the six-step sequence from the arriving action potential to a shortened sarcomere, one step per line and in order. Beside step two write which channel opens and where it sits. Beside step three write what opens RyR2 and from which side of the channel. Beside step four write the starting and peak cytosolic Ca2+ values. Beside step five write what troponin I and tropomyosin each do. Then write the sequence backwards as the relaxation sequence, naming what has to happen at troponin C and at each myosin head.

Rebuild from memory: Excitation-contraction coupling in a ventricular myocyte

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-05-01-C02-CP01

Skeletal against cardiac excitation-contraction coupling

Fill in every cell from memory, then open the notes and correct what you missed.

Feature	Skeletal muscle	Cardiac (ventricular) muscle
What excites the cell		
Where the T tubule sits along the sarcomere, and the name of the junction		
How the voltage sensor opens the sarcoplasmic reticulum release channel		
Whether extracellular Ca2+ is needed for a twitch		
Duration of the action potential		
Ca2+ sensor protein and how many Ca2+ binding sites it has		
What a Ca2+ free solution does to contraction within a minute		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-05-01-C03-BD1

Build the calcium removal table from memory

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C03 Activity: NPLEX-P1-CARD-05-01-C03-BD1 Date: _____

On blank paper build a table of the pathways that lower cytosolic Ca2+ in a ventricular myocyte, one row each for SERCA2a, the Na+/Ca2+ exchanger, and the plasma membrane Ca2+-ATPase with mitochondrial uptake. Use the columns location, energy source with the stoichiometry, approximate share of Ca2+ removed, and main regulator, and fill every cell. Underneath write the two values cytosolic Ca2+ moves between during relaxation. Then write three sentences on why relaxation costs ATP, naming what each myosin head needs in order to let go of actin. Finish by writing what unphosphorylated phospholamban does to SERCA2a, what phosphorylation of phospholamban does, and which two kinases phosphorylate it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-05-01-C03-BD2

Partial inhibition of the sodium pump, beat by beat

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C03 Activity: NPLEX-P1-CARD-05-01-C03-BD2 Date: _____

The Na+/K+-ATPase of a ventricular myocyte is partially inhibited. Work out what follows in six numbered steps, from the change in cytosolic Na+ through to the force of each beat, naming at every step which transporter is affected and why. Say which beat the force change appears on and why it takes several beats to settle. Then write what happens if the inhibition goes further: name the state the sarcoplasmic reticulum reaches, what it does in diastole, what the Na+/Ca2+ exchanger makes of that release, and the membrane event that results. Finish by writing why ischemia drives cytosolic Ca2+ up by the same route, naming the exchanger that adds Na+ during ischemia.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-05-01-C03-RT01

Rebuild calcium cycling in one heartbeat

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C03 Activity: NPLEX-P1-CARD-05-01-C03-RT01 Date: _____

Rebuild the six-step Ca2+ cycle of a single heartbeat, one step per line and in order, starting with trigger entry and ending with the relaxed cell. Beside each step write the structure doing the work and, where the notes gave one, the share of Ca2+ it handles. Then write the two balances that must hold for the cell to stay in a steady state, one line each. Underneath, write what happens to sarcoplasmic reticulum load and to the force of later beats if extrusion falls behind entry, and what happens if reuptake falls behind release.

Rebuild from memory: Ca2+ cycling in one heartbeat

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-05-01-C03-PR01

Predict what happens when coronary flow stops

A coronary artery is suddenly occluded and the myocytes it supplies lose oxidative phosphorylation. Predict each item below, write the direction of the change, give the time course where the notes gave one, and name in one line the molecule or transporter that explains it.

What changes	Up, down, or no change	Why, in one or two lines
What happens to creatine phosphate and to the products of anaerobic glycolysis		
What happens to the Ca2+ sensitivity of troponin C and to force per cross-bridge, and how soon		
What happens to cytosolic Na+ and to the driving force for the Na+/Ca2+ exchanger		
What happens to the rate of SERCA2a pumping and to diastolic cytosolic Ca2+		
What happens to ventricular stiffness and to end-diastolic pressure		
Which of contraction and relaxation is impaired first, and where that sits in the order of findings in the ischemic cascade		
What happens inside the cell if the occlusion lasts 20 to 40 minutes, and what Ca2+ overload does on reperfusion		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-05-01-C04-BD1

Draw the beta-1 cascade and its targets

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C04 Activity: NPLEX-P1-CARD-05-01-C04-BD1 Date: _____

Draw the beta-1 adrenergic cascade in a ventricular myocyte from the catecholamine arriving at the receptor to the change in the Ca2+ transient, naming the G protein and its alpha subunit, the enzyme it stimulates, the second messenger, the kinase, and the protein that holds the kinase near its targets. Name the two catecholamines and where each comes from. Beside the drawing build a table of protein kinase A targets with one row each for the L-type Ca2+ channel, phospholamban, troponin I, myosin-binding protein C, and the IKs channel, with columns for what phosphorylation does to the protein and what that means for the beat. Finish by writing the three things that end the signal.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-05-01-C04-BD2

Take one target out of the cascade

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C04 Activity: NPLEX-P1-CARD-05-01-C04-BD2 Date: _____

A mouse expresses a mutant phospholamban that cannot be phosphorylated but still binds SERCA2a. It is given a beta-1 agonist. Write what still happens to heart rate, to the L-type Ca2+ current, and to the force of contraction, then write what does not happen, naming the two separate jobs the normal phospholamban response does within one beat and across successive beats. Write what filling looks like at a fast heart rate in this animal and why. Next, explain in three sentences why troponin I phosphorylation does not weaken the beat in a normal heart. Then write why a person who has lost IKs develops arrhythmias during exercise rather than at rest. Finish with chronic heart failure: list the four changes the notes give in the beta-1 pathway and its downstream proteins, and write the consequence of each for force, for relaxation, and for the response to exercise.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-05-01-C04-CA01

Breathlessness on exertion in a man with a dilated ventricle

A 61-year-old man comes to the physician because of 6 months of increasing breathlessness on exertion. He stops after one flight of stairs, where a year ago he walked up three. Temperature is 36.8 °C (98.2 °F), pulse is 104/min and regular, respirations are 20/min, and blood pressure is 112/78 mm Hg. Physical examination shows a displaced apical impulse, crackles at both lung bases, and jugular venous distension. Echocardiography shows a dilated left ventricle with reduced wall motion, a low ejection fraction, and slowed early diastolic filling. During a treadmill test his pulse rises only to 118/min and his stroke volume does not increase. Plasma norepinephrine is elevated. Myocardial biopsy from a research protocol shows reduced beta-1 receptor density, increased G protein-coupled receptor kinase 2, reduced SERCA2a, and phospholamban that is largely unphosphorylated.

1. Recognize. Name the signaling pathway that is failing in this myocardium and list the four biopsy findings that place the fault in it.

2. Explain. Explain how each of those four findings changes the Ca2+ transient, the force of contraction, and the speed of relaxation, naming the protein kinase A target involved in each.

3. Differentiate. Write how this picture differs from a heart whose contractility is normal but whose filling is simply low, naming what the Ca2+ transient is doing in each and what the ventricular function curve does in each.

4. Investigate. Write which measurements on the exercise test and the echocardiogram you would use to show that the fault is in the response to catecholamines rather than in filling, and what each measurement would show.

5. Decide. State which single cellular lever is most responsible for the slowed early diastolic filling, and what that does to filling once the heart rate rises.

6. Justify. Justify that answer from the share of activator Ca2+ that SERCA2a normally removes, what unphosphorylated phospholamban does to the pump, and what a smaller sarcoplasmic reticulum store then does to the force of later beats.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-05-01-C05-BD1

Draw the length-tension curves

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C05 Activity: NPLEX-P1-CARD-05-01-C05-BD1 Date: _____

Draw sarcomere length on the horizontal axis and tension on the vertical axis, and plot both an active tension curve and a passive tension curve for cardiac muscle. Mark the thick filament length, the thin filament length, the sarcomere length at which overlap is optimal, the length at which active force reaches zero on the descending limb, and the range ventricular sarcomeres actually work over from low filling to high filling. Write what limits force on the ascending limb and what limits it on the descending limb, one line each. Then write what length-dependent activation is, which way it shifts the force versus Ca2+ curve, and the three mechanisms the notes propose for it. Finish by writing which two tissue components make passive tension climb steeply above 2.2 µm.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-05-01-C05-BD2

Tell a preload change from an inotropic change

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-5 Cardiac Muscle Contraction Competency: NPLEX-P1-CARD-05-01-C05 Activity: NPLEX-P1-CARD-05-01-C05-BD2 Date: _____

Write the two ways skeletal muscle grades its force, then write why the heart can use neither, naming the property of the myocyte network that rules out one and the property of the action potential that rules out the other. List the three levers the heart uses instead and, for each, give one thing that raises it and one thing that lowers it. Write the one condition about troponin C that has to hold for these levers to work at all. Next, two hearts each raise their stroke volume. In the first the sarcomeres lengthen with almost no change in the Ca2+ transient; in the second the Ca2+ transient grows. For each write what happens on the ventricular function curve, whether force rises at a given sarcomere length, and which of the two a beta-1 agonist produces. Finish by writing why a dilated failing ventricle is weak even though it is stretched, naming the three reasons the notes give and the structural change that keeps its sarcomeres off the descending limb.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-05-01-C05-CP01

Length-tension and force grading in skeletal against cardiac muscle

Fill in every cell from memory, then open the notes and correct what you missed.

Feature	Skeletal muscle	Cardiac muscle
Sarcomere length the muscle normally works at		
What the active length-tension curve rests on		
Passive tension at long sarcomere lengths		
Whether recruitment and summation are available, and why		
How large a part Ca2+ delivery and myofilament Ca2+ sensitivity play in grading force		
How much starting length matters in normal function, and the named mechanism in the heart		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-05-01-C01-Q01

A widened QRS with a high plasma potassium

A 58-year-old woman with chronic kidney disease is brought to the emergency department because of weakness. Temperature is 36.9 °C (98.4 °F), pulse is 58/min, respirations are 16/min, and blood pressure is 104/68 mm Hg. Serum studies show a K+ of 8.1 mmol/L (reference range 3.5 to 5.0 mmol/L). The electrocardiogram shows peaked T waves and a QRS duration of 0.16 s. Which of the following best explains the widened QRS complex?

A. Increased conductance of the rapid delayed rectifier K+ current

B. Prolonged opening of L-type Ca2+ channels during the plateau

C. A less negative resting potential leaving fast Na+ channels inactivated

D. Reduced Ca2+ release from the sarcoplasmic reticulum through RyR2

Why did you pick it?

Board-style question NPLEX-P1-CARD-05-01-C02-Q02

Two muscles in a calcium free bath

In an experiment, an isolated perfused mammalian heart and an isolated skeletal muscle preparation are switched to a Ca2+ free solution and stimulated at a constant rate. Within 1 minute the heart develops almost no force, while the skeletal muscle continues to twitch with little change. Which of the following best explains the difference between the two preparations?

A. Cardiac release channels open only in response to Ca2+ entering through L-type channels, while the skeletal voltage sensor opens its release channel mechanically

B. Cardiac myocytes use calmodulin and myosin light chain kinase, while skeletal fibers use troponin C

C. Cardiac sarcoplasmic reticulum holds no Ca2+ store, so all activator Ca2+ enters from outside the cell

D. Cardiac T tubules lie at the A-I junction, which places L-type channels too far from the release channels

Why did you pick it?

Board-style question NPLEX-P1-CARD-05-01-C05-Q03

Force doubles with a small change in overlap

A strip of ventricular muscle is stretched from a sarcomere length of 1.9 µm to 2.2 µm and stimulated at a fixed interval. Active force roughly doubles, while the calculated change in thick and thin filament overlap over this range is modest and the peak cytosolic Ca2+ of each twitch is unchanged. Which of the following best accounts for most of the increase in force?

A. Recruitment of additional myocytes that were not contracting at the shorter length

B. Summation of successive twitches as the refractory period shortens with stretch

C. A larger trigger Ca2+ influx through L-type Ca2+ channels at the longer length

D. Greater myofilament Ca2+ sensitivity at the longer sarcomere length

Why did you pick it?